

Splunk SIEM Lab Report

SOC Tier 1 Training — SSH Log Ingestion & SPL Analysis
Analyst: John | Date: 2026-09-11 | Environment: Kali Linux VM, Splunk Enterprise 10.4.3

1. Objective

As a follow-up to command-line log analysis (grep/awk) performed on a simulated SSH authentication log, Splunk Enterprise was installed locally to reproduce the same triage workflow inside a real SIEM platform. The goal was to build hands-on familiarity with Splunk installation, data ingestion, and the Search Processing Language (SPL) used daily by SOC analysts.

2. Environment Setup

Splunk Enterprise (version 10.4.3) was downloaded and installed via .deb package on a Kali Linux virtual machine. The Splunk daemon was started with root privileges enabled explicitly, and an administrator account was created for access to Splunk Web.

- Installed via: `sudo dpkg -i splunk-10.4.3-linux-amd64.deb`
- Started via: `sudo /opt/splunk/bin/splunk start --run-as-root --accept-license`
- Web interface accessed at: `http://127.0.0.1:8000`

3. Data Ingestion

The sample `auth.log` file (previously used for command-line triage) was ingested into Splunk's default "main" index using the CLI oneshot method, which indexes a single file directly without requiring the web upload wizard:

```
sudo /opt/splunk/bin/splunk add oneshot /home/bassbi4422/soc-lesson/auth.log -index main --run-as-root
```

Splunk automatically classified the data with sourcetype `linux_secure`, its built-in recognition pattern for authentication logs, while retaining `source=auth.log` as the original filename reference.

4. SPL Queries and Results

The following searches were run in Splunk's Search & Reporting app to identify failed SSH login attempts and the source IPs responsible, mirroring the earlier grep/awk pipeline:

Query 1 — Isolate failed login events:

```
source="auth.log" "Failed password"
```

Result: 5 matching events, confirming the same count found via the command line.

Query 2 — Extract source IP and count by IP:

```
source="auth.log" "Failed password" | rex "from (?<src_ip>\d+\.\d+\.\d+\.\d+)" | stats count by src_ip
```

Source IP	Failed Attempts
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185.220.101.4	3
45.155.205.77	2

This result exactly matches the findings from the earlier command-line analysis (`grep | awk | sort | uniq -c`), confirming consistent results across both methods.

5. Key Takeaways

- Successfully installed and configured a SIEM platform end-to-end on a self-managed VM.
- Learned the distinction between Splunk's `source` and `sourcetype` fields, and how automatic `sourcetype` classification can affect search results.
- Practiced core SPL syntax: field extraction with `rex` and aggregation with `stats count by`.
- Validated findings by reproducing the same security triage result using two independent methods (Linux command line and Splunk SPL), reinforcing confidence in the analysis.

This report documents a self-directed SOC Tier 1 training exercise focused on SIEM tool proficiency using Splunk Enterprise on a local Kali Linux lab environment.